

Temporal Domain Generalization + Kalman Adaptation Project

This project contains code for constructing MIMIC-IV temporal domain generalization datasets, analyzing temporal drift, and testing Kalman-guided predictive adaptation.

Recommended workflow

1. Create datasets

Run `dataset_creation.py` first. It expects the following MIMIC-IV files under:

```
~/scratch/mimiciv_wildtime_data/
```

Required files:

```
patients.csv.gz
admissions.csv.gz
diagnoses_icd.csv.gz
procedures_icd.csv.gz
```

Command:

```
python scripts/dataset_creation.py
```

By default, it saves outputs to:

```
~/scratch/temporal_dg_outputs_noleak/
```

It generates three datasets:

<code>dataset_a_demo_dx_nocad.csv</code>	# demographics + non-CAD diagnosis features
<code>dataset_b_demo_dx_proc_nocad.csv</code>	# demographics + non-CAD diagnosis features + procedures
<code>dataset_c_demo_proc.csv</code>	# demographics + procedures only

The CAD label is created from ICD-10 I20-I25 and ICD-9 410-414 codes. CAD-defining diagnosis codes are removed from diagnosis features to reduce direct label leakage.

2. Analyze temporal drift/generalization

For one dataset, especially Dataset B:

```
python scripts/datasetb_analysis.py \
  --dataset_path ~/scratch/temporal_dg_outputs_noleak/dataset_b_demo_dx_proc_nocad.csv \
  --out_dir outputs/dataset_b_analysis
```

For all `dataset_*.csv` files in a directory:

```
python scripts/temporal_dg_analysis.py \
  --input_dir ~/scratch/temporal_dg_outputs_noleak \
  --out_dir outputs/temporal_analysis
```

Note: `temporal_dg_analysis.py` currently searches for files matching `dataset_*.csv`. If your generated files are named `dataset_a_demo_dx_nocad.csv`, etc., this works.

3. Run Kalman-guided adaptation

```
python scripts/kalman_initial.py \  
  --dataset_path ~/scratch/temporal_dg_outputs_noleak/dataset_b_demo_dx_proc_nocad.csv \  
  --out_dir outputs/kalman_dataset_b \  
  --device cpu
```

If using a GPU on the cluster:

```
python scripts/kalman_initial.py \  
  --dataset_path ~/scratch/temporal_dg_outputs_noleak/dataset_b_demo_dx_proc_nocad.csv \  
  --out_dir outputs/kalman_dataset_b \  
  --device cuda
```

Main scripts

<code>scripts/dataset_creation.py</code>	Build datasets A/B/C from raw MIMIC-IV tables
<code>scripts/datasetb_analysis.py</code>	Single-dataset temporal generalization and drift analysis
<code>scripts/temporal_dg_analysis.py</code>	Multi-dataset temporal generalization and drift analysis
<code>scripts/kalman_initial.py</code>	Kalman-guided predictive adaptation experiment

Removed/archived

`pcavisualization.py` was intentionally removed because it was a one-off PCA visualization script with hard-coded paths and an older label convention.

Dependencies

Install with:

```
pip install -r requirements.txt
```

Key dependencies:

```
numpy  
pandas  
matplotlib  
scikit-learn  
scipy  
pykalman  
torch
```